12-WEEK AI ENGINEER LEARNING PLAN

Curriculum Roadmap, Weekly Topics, Hands-On Projects & Curated Learning Resources

Curriculum Overview: This 12-week roadmap progresses from core software engineering, Python tooling, and databases to cutting-edge AI systems including Fine-Tuning, RAG, FastAPI backend engineering, Docker deployment, Production Agent Memory, Model Context Protocol (MCP), and Voice LLMOps. Click any of the highlighted blue resource links throughout this guide to open the corresponding video, course, or GitHub code repository.

PHASE 1: PROGRAMMING & SOFTWARE FOUNDATIONS

| WEEK 1 — Core Python, Tooling & News Summarizer

Key Concepts: Variables, loops, data structures, classes, imports, HTTP requests/responses, and JSON data handling.

Project 1: News Summarizer — Read news articles from a CSV using pandas, make OpenAI API calls to summarize them, write HTTP/JSON handlers, and add unit/integration tests.

Learning Resource / Topic	Link / Access URL
Python for AI - Dave Ebbelaar	Python for AI - Dave Ebbelaar
uv Package Manager - Corey Schafer	uv Package Manager - Corey Schafer
GitHub Hello World Guide	GitHub Hello World Guide
OpenAI API Quickstart Documentation	OpenAI API Quickstart Documentation
Unit Testing in Python (Pytest) - pixegami	Unit Testing in Python (Pytest) - pixegami
API Integration Testing with PyTest - pixegami	API Integration Testing with PyTest - pixegami

| WEEK 2 — Testing, Continuous Integration & Optional ML

Key Concepts: Continuous Integration (CI) test execution on push/pull requests, plus key ML terminology (regression, decision trees, random forests, overfitting/underfitting, bias-variance trade-off, precision/recall).

Learning Resource / Topic	Link / Access URL
GitHub Skills - Test with Actions	GitHub Skills - Test with Actions
Microsoft ML for Beginners (Optional)	Microsoft ML for Beginners (Optional)

PHASE 2: DATABASES & LLM FUNDAMENTALS

| WEEK 3 — SQL & Database Integration

Key Concepts: SQL selection, filtering, joins, grouping, and relational schema design.

Project 1 Upgrade: Replace CSV storage by setting up a local **PostgreSQL database** to store and query news summaries directly in Python.

Learning Resource / Topic	Link / Access URL
SQLBolt Interactive Tutorial	SQLBolt Interactive Tutorial

| WEEK 4 — LLM Mechanics & Architecture

Key Concepts: Neural network weights, training vs. inference, backpropagation, transformer architecture, tokenizers, vector embeddings, and context windows.

Learning Resource / Topic	Link / Access URL
Intro to Large Language Models - Andrej Karpathy	Intro to Large Language Models - Andrej Karpathy
Zero to Hero Neural Networks (Optional) - Karpathy	Zero to Hero Neural Networks (Optional) - Karpathy
Hugging Face LLM Course (Chapter 1)	Hugging Face LLM Course (Chapter 1)

PHASE 3: FINE-TUNING, RAG & API ENGINEERING

| WEEK 5 — Parameter-Efficient Fine-Tuning & RAG Intro

Key Concepts: Parameter-efficient fine-tuning using **LoRA**, evaluating prompting vs. context vs. fine-tuning, and Retrieval-Augmented Generation (document chunking, vector search with pgvector, hybrid search, and evaluation).

Learning Resource / Topic	Link / Access URL
Unsloth Notebooks Catalog (Fine-Tuning)	Unsloth Notebooks Catalog (Fine-Tuning)

| WEEK 6 — Basic RAG Application (Project 2)

Key Concepts: Document ingestion, text splitting, generating embeddings, hybrid retrieval (combining keyword and vector search), and answer synthesis with sources.

Project 2: Streamlit RAG App — Construct a Streamlit application backed by local vector search and PostgreSQL.

Learning Resource / Topic	Link / Access URL
Beginner Local RAG Code Repository	Beginner Local RAG Code Repository
Build a Local RAG System (Guide Part 1)	Build a Local RAG System (Guide Part 1)
Build a Local RAG System (Guide Part 2)	Build a Local RAG System (Guide Part 2)

| WEEK 7 — API Engineering with FastAPI

Key Concepts: Async programming, request validation, database migrations, routers, authentication, authorization, dependency injection, and streaming error states.

Project 2 Upgrade: Refactor your RAG app to use a **FastAPI backend** with user authentication, access control, and asynchronous endpoints.

Learning Resource / Topic	Link / Access URL
FastAPI Tutorials Series - Corey Schafer	FastAPI Tutorials Series - Corey Schafer
FastAPI Dependencies Tutorial	FastAPI Dependencies Tutorial

PHASE 4: CONTAINERIZATION, DEPLOYMENT & CD

| WEEK 8 — Containerization & Cloud Deployment

Key Concepts: Docker images, containers, Dockerfiles, Compose, environment secrets management, logging, health checks, timeouts, and cloud cost controls.

Learning Resource / Topic	Link / Access URL
Docker Tutorial for Beginners - TechWorld with Nana	Docker Tutorial for Beginners - TechWorld with Nana
Deploy FastAPI with Docker to Cloud Run - Corey Schafer	Deploy FastAPI with Docker to Cloud Run - Corey Schafer

| WEEK 9 — Continuous Delivery & Production Agent (Project 3)

Key Concepts: Continuous Delivery (CD) release pipelines, automated AI evaluation checks, and production agent architectures.

Project 3: Production Agentic System — Build a production-grade agentic system incorporating engineering standards, tool usage, and release verification.

Learning Resource / Topic	Link / Access URL
Production Agentic RAG Course Repository	Production Agentic RAG Course Repository
How a Production AI Agent System Actually Works	How a Production AI Agent System Actually Works

PHASE 5: AGENT MEMORY, MCP & SYSTEM DESIGN

| WEEK 10 — Agent Memory Architectures

Key Concepts: Managing short-term conversation context and long-term user memory state across sessions.

Learning Resource / Topic	Link / Access URL
AI Agent Memory: 7 Types Explained	AI Agent Memory: 7 Types Explained
Memory for Agents - LangChain Conceptual Overview	Memory for Agents - LangChain Conceptual Overview
Long-Term Memory Agent Template - LangChain	Long-Term Memory Agent Template - LangChain

| WEEK 11 — Model Context Protocol (MCP) & System Design

Key Concepts: Standardizing tool and data server connectivity with **MCP**, plus AI system design patterns (response caching, rate-limiting, request queuing).

Learning Resource / Topic	Link / Access URL
Intro to MCP Servers with Python - freeCodeCamp	Intro to MCP Servers with Python - freeCodeCamp
Hugging Face MCP Course	Hugging Face MCP Course
AI System Design: 7 Patterns Explained	AI System Design: 7 Patterns Explained

PHASE 6: VOICE AGENTS & LLMOPS

| WEEK 12 — Voice Agent & Observability (Project 4)

Key Concepts: Real-time audio interaction, speech-to-text / text-to-speech pipelines, and LLMOps monitoring (latency, cost tracking, prompt tracing, error metrics).

Project 4: Observable Voice Agent — Build an interactive voice job search agent equipped with full operational observability.

Learning Resource / Topic	Link / Access URL
Observable Job Agent Repository	Observable Job Agent Repository